NAVONILA KARMAKAR

Final Year BS-MS Physics Student
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RESEARCH EXPERIENCE

MS Thesis Project at **Max Planck Institute for Astrophysics, Germany** *06/2026-Present*
Supervisors: **Prof. Dr. Volker Springel and Dr. Alankar Dutta**
Hydrodynamic cloud-crushing simulation using PLUTO: Extending classical cloud-crushing (cold gas clouds of CGM in hot galactic wind) setup in two ways: (1) adding wind divergence/expansion, and (2) fractal, multi-clump cloud complex instead of single spherical cloud, and analysing how each affects the cloud survival, evolution and morphology, and the physics behind, ongoing work on cloud analysis. (Presentation)

Independent Study at **IISER Kolkata**; Supervisor: **Prof. Rajesh Kumble Nayak** *08/2026-Present*
Semester project on **Bayesian inference and parameter estimation of gravitational waves**: basics of Bayesian inference and parameter estimation, compact binary coalescence parameter estimation using Bilby, ongoing work on GW250114.

Remote Project Intern at **ESO, Germany**; Supervisor: **Dr. Morten Andersen** *05/2025-Present*
Characterizing the intermediate and low-mass stellar content of Barbá 2 cluster: imaging data of H and Ks bands from Gemini FLAMINGOS2, data reduction using DRAGONS pipreline and Astropy, infrared photometric (aperture) study of cluster stars using photutils, calibration with 2MASS photometric data, completeness limit determination from radially binned CMD and corresponding luminosity function, estimation of stellar masses from individual dereddening, and finally derivation of cluster mass and stellar number density profile.

Astronomy Lab Course Semester Project, **IISER Kolkata** *04/2026-05/2026*
Structural and Morphological Analysis of Galaxies using Image Statistics: Data acquisition from SDSS SkyServer for three different types of galaxies: NGC 2967 (Sc spiral), NGC 3351 (SBb barred spiral), NGC 3379 (E1 elliptical), image processing, surface brightness profile and model fitting, error and noise analysis, CAS morphological parameters determination, isophote analysis, global fourier analysis (2D FFT), radial Fourier analysis (Annular 1D FFT), correlation and statistical analysis, radial structural decomposition, final comparison and classification. (Report)

LIGO GET-GO Summer Intern at **ICG, University of Portsmouth, UK** *05/2025-08/2025*
Supervisors: **Prof. Dr. Laura Nuttall and Dr. Rahul Dhurkunde**
Mitigating the effect of glitches in the search of precessing systems: basics of GW searches such as matched filtering, signal consistency tests, coincident detection, ranking statistic, candidate significance etc using PyCBC; differences in waveform, template bank and ranking statistic for precessing searches; mathematical formalism of harmonic decomposition of precessing waveforms, performing precessing searches on a few days of data of O3 in CIT cluster, identification of GPS times of glitches that are affecting precessing searches, types of glitches, estimation of sensitivity for a simulated population of signals (injections) at a given FAR, comparing the sensitivity of the new model with existing general searches. (Presentation)

Summer Research Fellow at **ARIES Nainital, India**; Supervisor: **Dr. Saurabh** *05/2024-07/2024*
Understanding Star Formation in Young Clusters from Ground-based Observation: observation in 3.6 m Devasthal Optical Telescope, imaging and spectroscopy with TANSPEC; data reduction using IRAF, UVBRI photometric (PSF) study of NGC 2316 stars using DAOPHOT II, calibration with Landolt's standard SA 98 to determine true magnitude of stars, 2MASS near-infrared photometric data

study for cluster identification, kinematic study from GAIA proper motion data for determining membership probability, astrometry using IRAF, GAIA Tool & WCSTools, isochrone fitting with intrinsic Zero-Age Main Sequence on HR Diagram of cluster stars for morphological analysis. (Report)

Summer Intern at **CESSI, IISER Kolkata**; Supervisor: **Dr. Sudip Kumar Garain** *07/2023*
Literature review on **Stellar Evolution: Interstellar Cloud to Black Holes**. (Report)

EDUCATION

Indian Institute of Science Education and Research (IISER), Kolkata *10/2022 - Present*
BS-MS Dual Degree, Major in Physics CGPA: 8.0
Coursework: classical mechanics, quantum mechanics, mathematical physics, thermodynamics and statistical physics, electrodynamics, computational physics, electronics theory & lab, nuclear physics lab, introductory astrophysics, general relativity, quantum field theory, condensed matter theory & lab, introductory cosmology, astronomy lab, fluid dynamics and MHD, atomic & optical physics, advanced optics theory & lab.

Bishnupur High School *08/2020 - 06/2022*
West Bengal Council of Higher Secondary Education(WBCHSE) Percentage: 93.8

Shibdas Central Girls' High School *01/2014 - 07/2020*
West Bengal Board of Secondary Education(WBBSE) Percentage: 93.8

PUBLICATIONS

Navonila Karmakar, Alankar Dutta. Fragmented in the flow: survival of fractal cloud complexes in expanding galactic outflows (Manuscript under preparation)

M. Andersen, **N. Karmakar**, S. Millstone. The infrared view of the Galactic star cluster Barba 2 (Submitted to Astronomy & Astrophysics in October 2026)

SKILLS

Programming Languages and Softwares: Python (SciPy, AstroPy, Photutils, h5py, PyCBC, Bilby), C/C++, Julia, MATLAB, LaTeX, IRAF, DS9, DAOPHOT II, GAIA Tool, OriginLab, Anaconda, Git, PLUTO, ParaView, GNUPlot, MESA, Phlegethon, CASA, HEASoft.

Languages: English, Bengali, Hindi

SUMMER & WINTER SCHOOLS/WORKSHOPS/CONFERENCES

IMPRS-HD Summer School, University of Heidelberg *09/2026*
Cosmic Multiplicity: Evolution and Fate of Binary Stars

PLUTO/OpenGadget3 Symposium *06/2026*
Max Planck Institute for Astronomy, Heidelberg, Germany

CMF2IMF - The Origin of the Stellar Initial Mass Function *06/2026*
European Southern Observatory, Garching, Germany

Cosmology, Astrophysics & Astronomy *03/2026*

Special Lecture Series by International Leading Experts, Kolkata

Beyond LIGO-India: Deci-Hz GW Space Mission, IISER Kolkata *03/2026*

Glorious Women Online Symposium, India (Online) *03/2026*

LIGO GET-GO Student Conference, University of Glagow, UK (Presentation) *07/2025*

SOKENDAI Asian Winter School, NAOJ & SOKENDAI, Japan *02/2025*

Sagan Summer Workshop, NASA Exoplanet Science Institute, Caltech *07/2024*

Lecture Series on **Physics of Inflation & Reheating**, **Raman Research Institute** *07/2024*

Summer School on Gravitational-Wave Astronomy, **ICTS** Bangalore *07/2024*

REYES student (virtual), **University of California** *06/2024-07/2024*

Summer School in Astronomy & Astrophysics, **IUCAA** Pune *05/2024-06/2024*

Workshop on Gravity, Cosmology and Raychaudhuri's Equation, IUCAA & JU *03/2024*

18th Kavli Asian Winter School on Strings, Particles and Cosmology *12/2023*
Yukawa Institute for Theoretical Physics and ICTP (online attendee)

10th IAPT National Student Symposium on Physics, Chandigarh *08/2023*

START (Space Science and Technology Awareness Training), **ISRO** *07/2023-08/2023*

HONOURS AND AWARDS

MPG-IISER MS Project Research Fellow *06/2026-Present*
Fellowship sponsored by **Max Planck Institute for Astrophysics**, selected through **MPA Internship** programme and **MPG-IISER** programme; only 1 student from all over India has got this fellowship in astrophysics (3 students in physics).

LIGO GET-GO Summer Research Fellow *05/2025-08/2025*
Gravitational wave detector Expertise and Technology-GrOwth (GET-GO) fellowship sponsored by **University of Glasgow**, selected through **LIGO Summer Undergraduate Research Fellowship (SURF)** programme; only 4-5 students from all over India get this fellowship every year.

IASc-INSA-NASI Summer Research Fellow *05/2024-07/2024*
Summer Research Fellowship programme (SRFP) sponsored by **Indian Academy of Sciences** (Bangalore), **Indian National Science Academy** (New Delhi), **National Academy of Sciences, India** (Prayagraj); prestigious summer fellowship for undergraduates in India.

Silver Honour (Youth), IAAC (International Astronomy and Astrophysics Competition) *06/2023*

Bronze Medalist (Senior), National Level Astrophysics Quiz Competition *09/2021*
IAPT and A.P.J. Abdul Kalam National Council of Young Scientists